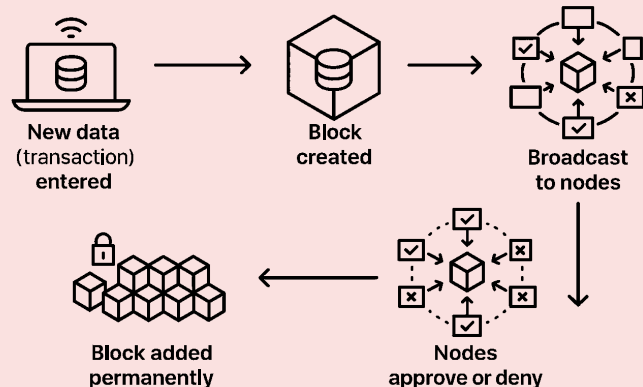


Blockchain

A distributed ledger technology that maintains a continuously growing list of records (blocks) linked and secured using cryptographic hashes, enabling trust-less transactions without intermediaries through consensus mechanisms.



Pros

- Immutable transactions
- Decentralized consensus
- Enhanced transparency

Cons

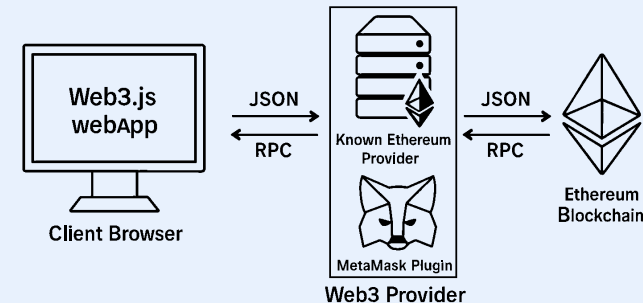
- Energy consumption
- Scalability limitations
- Transaction latency

Application

- Smart contracts
- Digital identity
- Supply-chain

Web3

A decentralized internet paradigm built on blockchain technology that enables user ownership of data, assets and digital identity through distributed protocols and peer-to-peer interactions without centralized intermediaries.



Pros

- Data ownership
- Censorship resistance
- Interoperability protocols

Cons

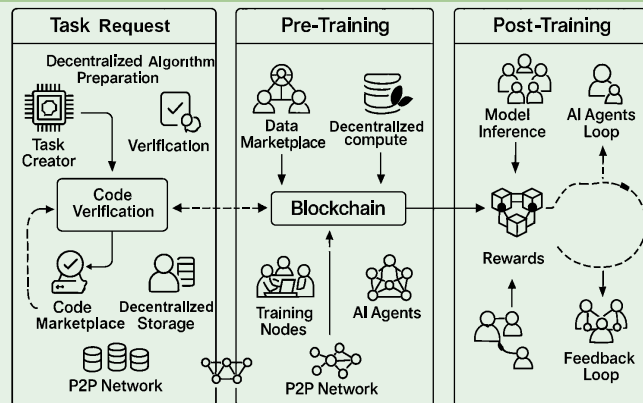
- Technical complexity
- User-experience
- Regulatory uncertainty

Application

- Decentralized finance
- Social networks
- NFT marketplaces

Decentralized AI

AI systems distributed across multiple nodes that enable collaborative learning and inference without centralized control, leveraging federated learning and multi-agent systems to preserve data privacy while maintaining model performance.



Pros

- Data-privacy
- Distributed computation
- Fault tolerance

Cons

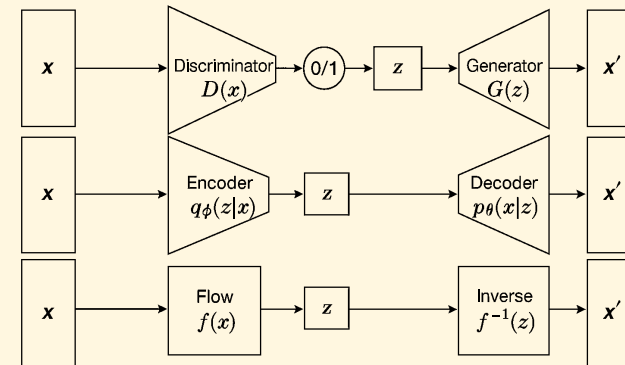
- Communication overhead
- Model consistency
- Coordination complexity

Application

- Federated learning
- Agentic systems
- Edge computing

Generative AI

AI systems capable of creating new content from existing data using advanced machine-learning techniques like transformers, GANs and diffusion models to generate text, images, code and multimedia content that mimics human creativity.



Pros

- Content generation
- Automated assistance
- Multimodal capabilities

Cons

- Computational requirements
- Hallucination issues
- Ethical concerns

Application

- Content creation
- Code generation
- Virtual assistants